

CHATEAUDUN AB, France

WMO: 071400

Lat: 48.0614N

Lon: 1.3764E

Elev: 433

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	25.6	17.5	13.5	25.5	21.0	16.0	27.8	30.1	49.2	27.4	48.7	5.3	40	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	88.3	67.8	84.5	66.4	80.8	65.3	70.1	83.5	68.2	80.3	66.6	77.5	8.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.0	73.9	64.0	90.8	72.3	62.5	86.0	70.5	34.2	83.7	32.7	80.4	31.4	77.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	21.6	19.2	16.7	95.7	6.0	3.9	12.4	98.5	8.9	100.8	5.5	103.0	1.2	105.9
			16.4	73.5	5.8	2.0	12.3	74.9	8.9	76.1	5.7	77.2	1.5	78.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	39.6	40.4	45.4	50.2	56.6	62.8	66.7	65.7	60.8	53.8	45.3	40.3	(6)
		DBStd	11.51	6.87	6.92	6.13	5.71	5.81	6.02	5.87	5.35	5.24	6.17	6.24	6.84	(7)
		HDD50	1347	326	271	169	66	10	0	0	0	34	164	307		(8)
		HDD65	4863	788	689	609	445	266	111	49	54	147	348	590	767	(9)
		CDD50	2213	3	3	25	71	216	385	517	488	324	152	24	5	(10)
		CDD65	251	0	0	0	0	6	46	100	77	21	1	0	0	(11)
		CDH74	2957	0	0	0	20	102	551	1154	855	256	19	0	0	(12)
		CDH80	965	0	0	0	1	11	174	417	299	61	2	0	0	(13)
(14)	Wind	WSAvg	9.1	10.1	10.4	10.1	9.2	8.4	8.2	8.5	8.0	8.1	8.5	9.5	10.1	(14)
(15)	Precipitation	PrecAvg	24.8	2.2	1.8	1.7	1.9	2.2	2.0	2.2	1.8	1.8	2.6	2.3	2.4	(15)
(16)		PrecMax	33.7	4.5	4.0	5.2	4.8	4.5	4.1	6.7	4.0	4.7	5.8	4.2	5.4	(16)
(17)		PrecMin	17.9	0.3	0.2	0.3	0.3	0.4	0.3	0.6	0.3	0.2	0.8	0.7	0.5	(17)
(18)		PrecStd	4.3	1.0	1.0	1.1	1.2	1.1	1.0	1.3	1.1	1.1	1.3	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.8	60.3	69.4	76.9	81.5	91.7	94.3	92.9	86.0	76.6	62.8	56.9		(19)
(20)		0.4% MCWB	53.4	51.0	54.0	60.0	66.5	71.1	69.0	68.8	64.9	63.3	58.5	54.4		(20)
(21)		2% DB	53.8	55.2	64.0	70.9	76.9	85.0	88.5	87.3	80.9	70.7	59.6	54.7		(21)
(22)		2% MCWB	51.8	50.2	53.1	57.5	63.4	67.9	67.7	67.0	64.3	61.6	56.8	52.6		(22)
(23)		5% DB	51.7	52.7	59.5	66.0	73.0	80.2	84.5	82.9	76.2	66.4	57.1	52.8		(23)
(24)		5% MCWB	49.7	49.2	51.6	55.3	61.7	65.8	66.3	65.8	62.8	59.5	54.4	50.9		(24)
(25)		10% DB	49.5	50.6	56.0	61.8	69.2	75.9	80.6	78.4	72.4	63.2	54.8	50.8		(25)
(26)		10% MCWB	47.7	47.9	50.4	52.9	59.5	63.9	65.3	64.0	61.2	58.1	52.4	49.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	54.3	53.1	56.5	61.9	68.7	73.3	71.8	71.3	68.3	65.3	60.0	55.1		(27)
(28)		0.4% MCDB	55.2	56.2	63.5	74.6	78.7	84.7	88.9	87.8	78.7	72.1	61.6	56.1		(28)
(29)		2% WB	52.1	51.3	54.6	58.4	65.1	70.1	69.7	68.9	66.2	62.7	57.2	53.1		(29)
(30)		2% MCDB	53.4	53.8	60.2	68.4	74.4	81.7	84.3	81.9	76.6	67.6	59.3	54.3		(30)
(31)		5% WB	50.0	49.8	52.9	56.1	62.6	67.3	67.9	67.1	64.3	60.9	54.8	51.2		(31)
(32)		5% MCDB	51.4	52.2	57.9	64.3	70.9	77.6	80.9	78.7	73.7	65.4	56.8	52.7		(32)
(33)		10% WB	47.8	48.0	51.1	53.8	60.3	65.0	66.2	65.5	62.6	59.1	52.6	49.2		(33)
(34)		10% MCDB	49.2	50.5	55.8	60.4	67.4	74.3	77.7	75.7	70.3	62.6	54.5	50.7		(34)
(35)	Mean Daily Temperature Range	MDBR	10.0	13.4	17.0	19.4	19.6	20.6	21.7	21.8	20.9	16.3	12.0	10.5		(35)
(36)		5% DB MCDBR	11.2	17.0	23.7	26.3	25.8	27.5	29.1	29.6	27.6	20.5	13.8	11.5		(36)
(37)		5% DB MCWBR	9.7	12.6	14.1	14.6	13.9	12.9	11.0	12.1	13.0	12.4	10.8	9.8		(37)
(38)		5% WB MCDBR	10.6	14.8	19.6	23.7	23.3	25.3	25.6	25.0	23.2	18.3	13.2	10.9		(38)
(39)		5% WB MCWBR	9.6	11.7	13.0	13.9	13.4	13.1	11.0	12.0	12.2	11.9	11.0	9.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.341	0.370	0.398	0.402	0.409	0.405	0.394	0.367	0.360	0.333	0.317		(40)
(41)		taud	2.435	2.382	2.297	2.224	2.254	2.270	2.275	2.325	2.385	2.414	2.439	2.409		(41)
(42)		Ebn at Noon	238	258	266	270	273	271	270	268	265	247	230	222		(42)
(43)		Edh at Noon	21	27	35	41	42	41	41	37	32	26	21	19		(43)
(44)	All-Sky Solar Radiation	RadAvg	312	568	941	1398	1645	1843	1765	1545	1200	708	379	268		(44)
(45)		RadStd	30	75	119	162	147	181	123	146	87	61	37	36		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page